SILO+ - Image Restoration via Attention and Multi-Scale Learning

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CS/ECE 8690 Computer Vision Final Project

Outline

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3. Approaches to the task
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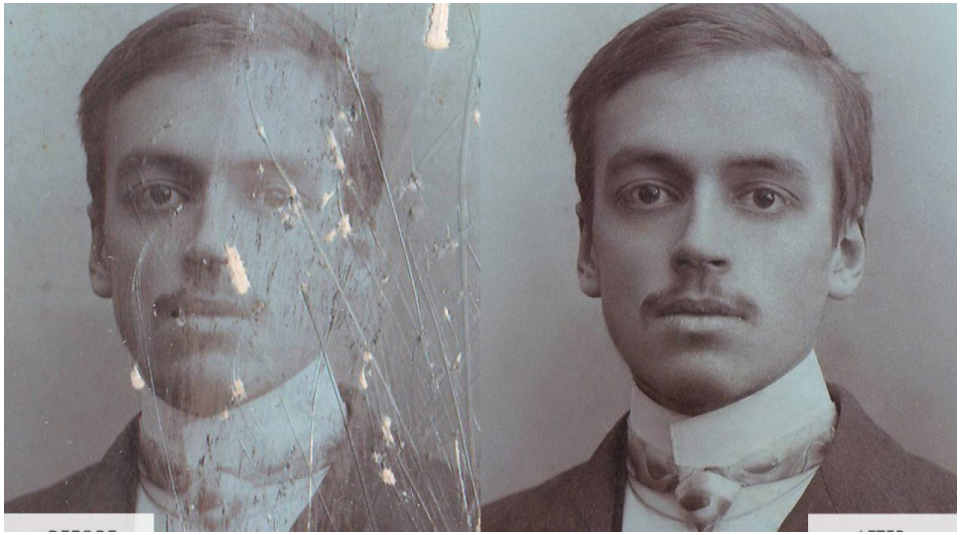
Motivation and Scope

- **Image restoration:** recover a clean image from a degraded measurement (inpainting, deblurring)

- **Why it matters:** photo restoration, medical/satellite imaging, surveillance recovery, content creation tools

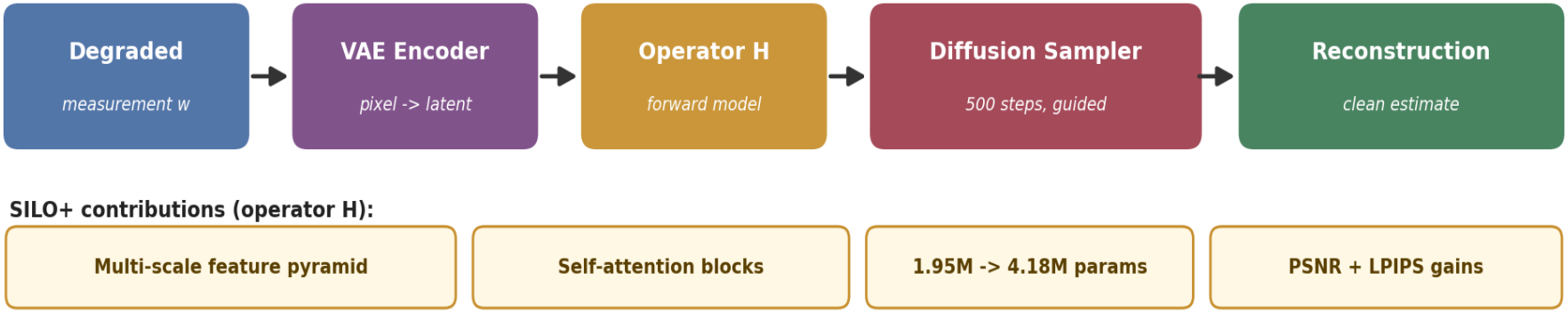


- **Existing approaches:** task-specific CNNs(DnCNN, deblur GANs), and recent diffusion-based solvers (DPS, DDRM,SILO)

- **Our scope:** build SILO and an enhanced SILO+ that adds attention and multi-scale learning, evaluate on FFHQ faces for inpainting and gaussian deblurring

Pipeline

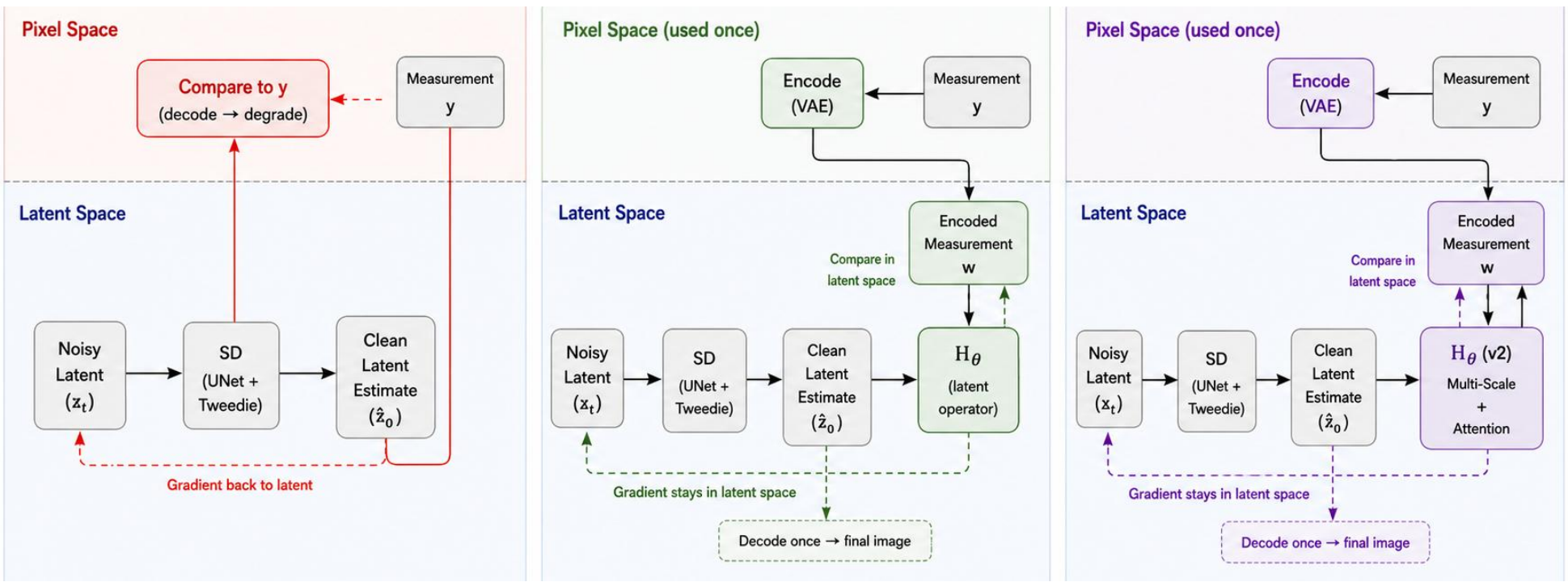
- **Backbone:** pretrained Stable Diffusion, VAE + UNet - frozen during training and inference
- **Trainable operator H_θ:** maps a clean latent to the latent of its degraded version, replacing the per-step decoder used by older methods
- **Inference:** encode measurement once → 500-step diffusion loop guided by $\|w - H_\theta(\hat{z}_0)\|$ → decode once at the end



Silo+ Pipeline: Latent Space Diffusion Guided Image Restoration

Approaches

- **Naive (LDPS-style):** decode every step → compare in pixel space → backprop through decoder. Slow and produces decoder-induced artifacts.
- **SILO:** decode only at thebend - A trained latent operator H_θ replaces the decoder inside the loop. 2.6 -10× faster, cleaner gradients.
- **SILO+:** same loop as SILO, with H_θ upgraded - adds spatial attention and multi-scale convolutions for better perceptual quality



Naïve Approach *Baseline SILO* *SILO+*

Algorithms and Module Implementation Details

- Modules: VAE encoder/decoder, frozen U-Net, learned operator H (baseline or +), DDIM sampler with measurement guidance
- Sampler loop: predict z_0_hat → compute residual r = w − H(z_0_hat) → step latent toward smaller r → repeat for 500 steps
- Stack: PyTorch, Diffusers 0.30, Transformers 4.44, PEFT, Accelerate, einops, LPIPS — trained on a single A100 (40 GB)
- Dataset: FFHQ-1000 (1,000 high-resolution face images at 512×512); 5-image held-out test set for evaluation
- Metrics: PSNR (pixel fidelity, higher is better) and LPIPS (perceptual similarity, lower is better)

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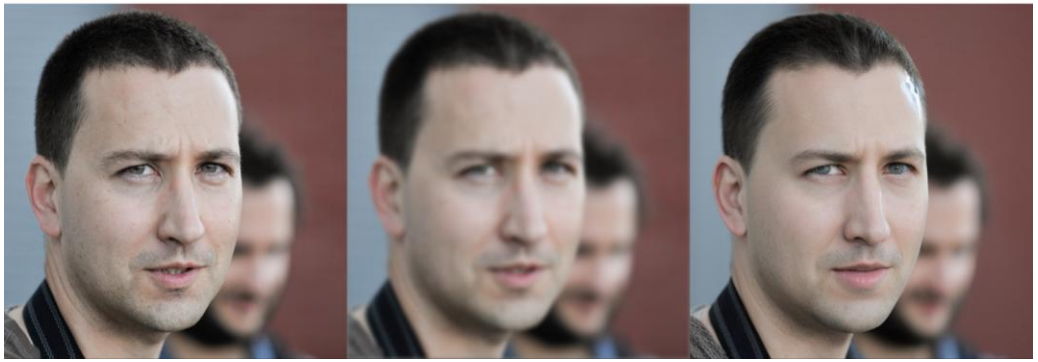
Measurement-Guided Diffusion

- Training: sample (z, t, ε); compute degraded latent; backprop $||H(z_0_pred) - w||$ through the operator only
- Inference: standard DDIM step from t→t−1, then a guidance correction toward smaller measurement residual
- Residual $||w - H(z_0_hat)||$ drops monotonically over the 500 steps (e.g., 99.3 → 21.6 for one inpaint test image)

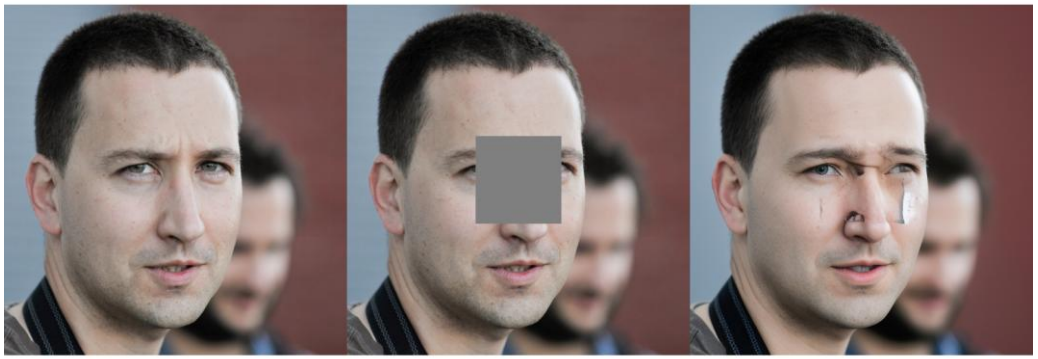
```
Sampling: measurement-guided DDIM (silo_v2.py)

# given:
# H        = trained operator (baseline or  +)
# w        = degraded measurement (in latent space)
# vae, unet, scheduler  = frozen Stable Diffusion components

z = randn(latent_shape)
for t in scheduler.timesteps:                    # 500 steps
    eps   = unet(z, t)                           # noise prediction
    z0_hat = (z − sigma_t · eps) / alpha_t        # predict clean latent
    z      = scheduler.step(eps, t, z)            # standard DDIM step
    z      = z − lr · grad || w − H(z0_hat) ||     # measurement guidance
image = vae.decode(z)                             # back to pixel space
```

Visual Results: Head-to-Head

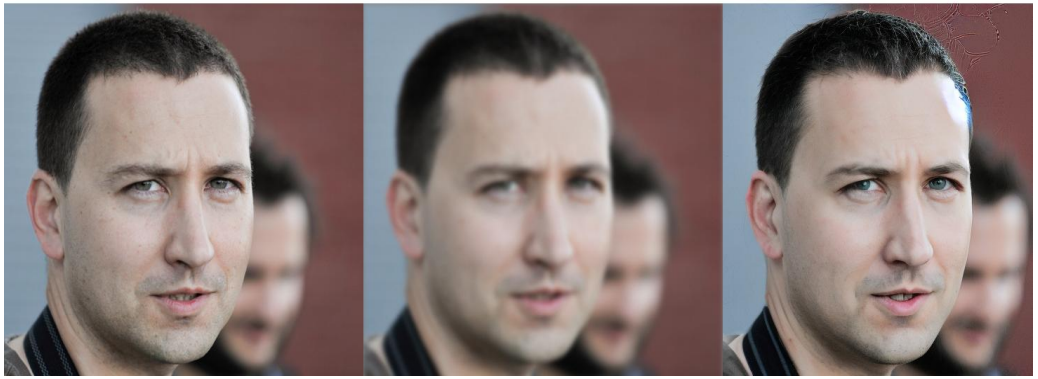
Gaussian Deblur (SILO)
Clean | Measurement | Reconstruction



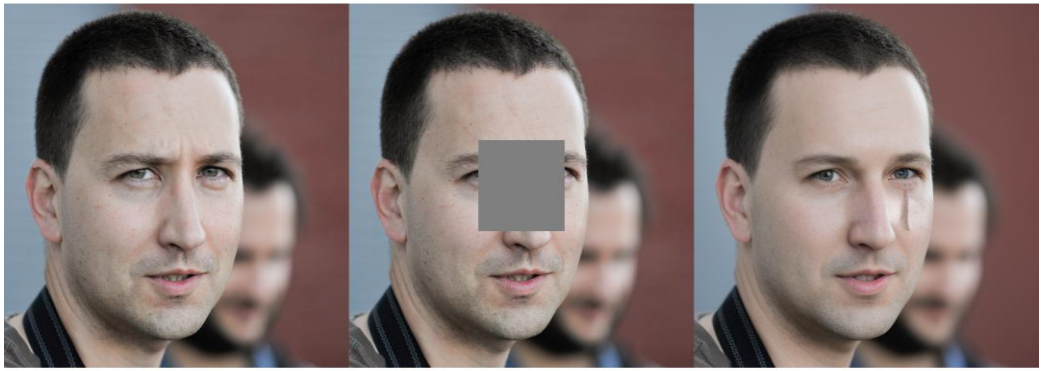
Inpainting (SILO)
Clean | Measurement | Reconstruction



Gaussian Deblur (SILO+)
Clean | Measurement | Reconstruction



Inpainting (SILO+)
Clean | Measurement | Reconstruction



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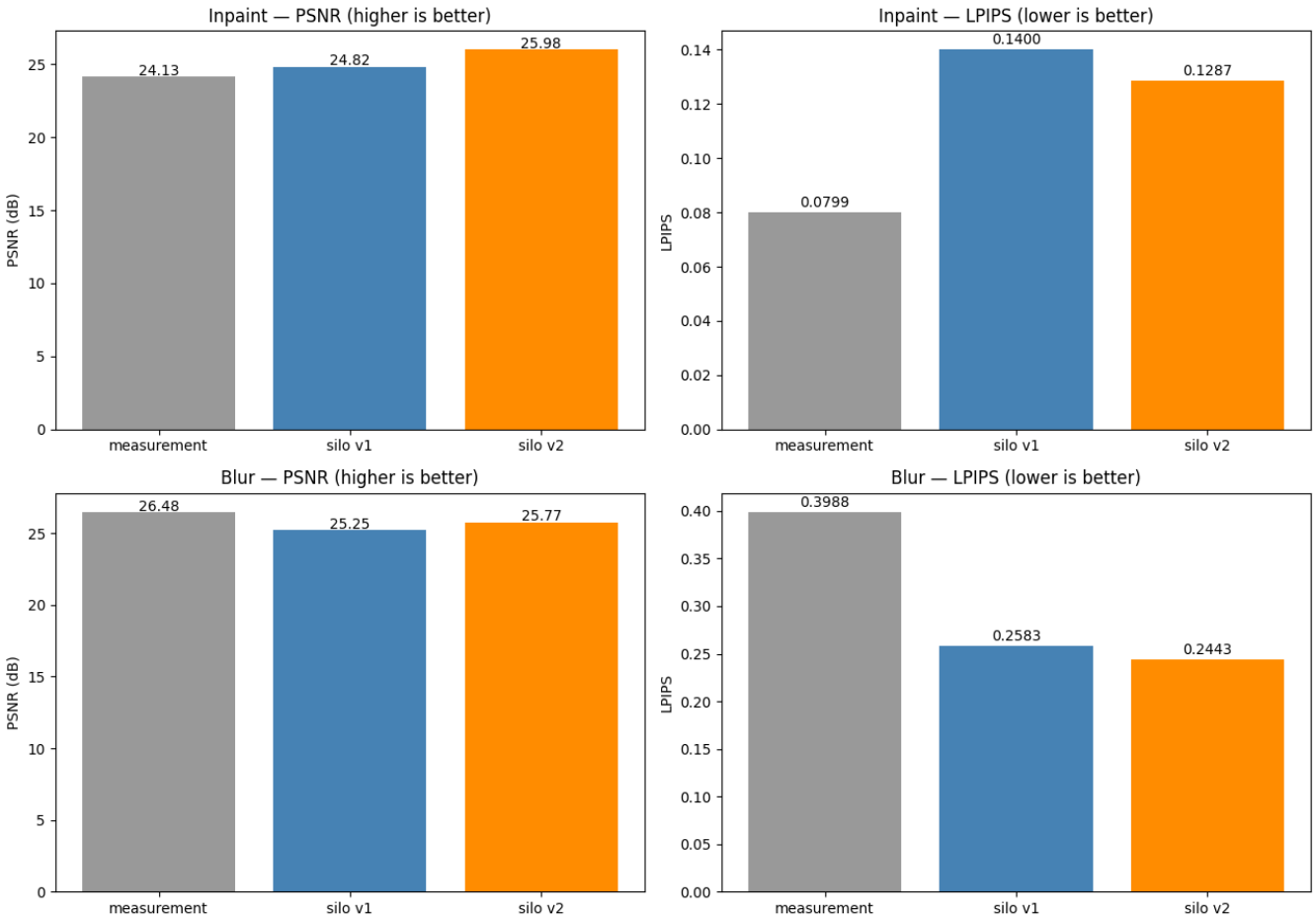
Quantitative Results

- Inpainting: SILO+ v2 wins both metrics - PSNR 25.98 dB, LPIPS 0.129 (+1.16 dB over v1)

- Deblurring: SILO+ v2 LPIPS 0.244 - 39% lower than blurred input (0.399)



Summary & Future Work

- **Goal vs. outcome:** built both SILO baseline and SILO+; achieved consistent LPIPS gains and a +1.16 dB inpainting PSNR lift

- **Problems encountered:** GPU memory pressure with the larger operator, slow 500-step sampling, residual artifacts on hard inpainting cases (e.g., eye-region masks)

- **Lessons learned:** PSNR alone is misleading - LPIPS captured perceptual gains that PSNR missed (esp. for deblurring)

- **Future work:** Implement readout-guidance H_θ (paper's approach, scale to FFHQ-train (~70k images), add super-resolution and JPEG decompression

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References

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Thank You

Questions, comments, ideas - we're ready

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